Keeping Alternatives Alive: A Comparison-Preserving Scaffold for Root Cause Analysis Under Closure Pressure

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Root cause analysis (RCA) in pharmaceutical manufacturing is designed to identify systemic causes — yet investigations routinely converge on individual blame before alternatives have been meaningfully compared. This pattern reflects a property of the investigative system rather than an individual reasoning failure: closure pressure and documentation requirements collapse the comparison space early, before systemic contributors can be meaningfully examined.

Drawing on Premature Accountability Convergence (PAC) — a framework describing how closure pressure accelerates the settling of responsibility during investigation — I ask the inverse design question: can AI expand interpretive space rather than compress it? I report a two-study empirical program with $N = 20$ GMP-experienced participants. Study 1 establishes that convergence is structural and time-sensitive: initial explanations were uniformly single-cause and person-centered, but broadened with additional reflection time. Study 2 introduces an AI-supported reasoning scaffold that functions as a comparison enforcer — not by suggesting causes, but by detecting premature convergence signals and triggering reopening prompts. The scaffold increased causal breadth, sustained alternative exploration, and reduced person-centered attribution. The most significant adoption barrier was organizational, revealing a tension between reasoning transparency and existing accountability practices.

I introduce comparison preservation as a design principle for AI-supported reasoning in accountability-critical environments, and argue that accountability in collaborative investigative work is structured before attribution — and that design can reopen the space where that structuring occurs.

Additional Key Words and Phrases: root cause analysis, accountability, closure pressure, human-AI interaction, comparison preservation, regulated environments

1 Introduction

Root cause analysis (RCA) in pharmaceutical manufacturing is a formal quality requirement embedded in collaborative investigative work: every deviation must be investigated, its cause documented, and corrective actions assigned before the record can be closed. In principle, investigation should compare multiple plausible causes before converging on an explanation. In practice, investigations unfold under closure pressure — temporal deadlines, documentation requirements, and institutional expectations that define when an investigation must be considered complete. Under these conditions, the first viable explanation tends to become the documented one, not because alternatives have been eliminated, but because the workflow does not keep them comparable long enough to be evaluated.

Prior work has largely framed premature convergence as a cognitive problem: investigators rely on recognition-primed decision making [4], anchor on initial hypotheses, or fail to generate sufficient alternatives. I argue this framing misidentifies the locus of failure. Premature convergence in regulated investigations is not primarily a property of how individuals reason — it is a property of how investigative workflows organize, interrupt, and foreclose comparison under institutional pressure.

This paper introduces *comparison preservation* as a design principle for AI-supported reasoning in accountability-critical environments. Drawing on Premature Accountability Convergence (PAC), I ask: rather than accelerating convergence, can AI be designed to expand and maintain the interpretive space within which comparison occurs?

This paper makes three contributions. First, it reframes premature convergence as a failure of investigative system design rather than individual cognition. Second, it introduces the *comparison enforcer* as a design role for AI in

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collaborative investigative settings — distinct from suggestion engines or automated classifiers. Third, it identifies organizational transparency as an adoption barrier specific to regulated environments, where making reasoning visible surfaces accountability conditions that existing practice leaves invisible.

2 Related Work

Vaughan's account of the Challenger launch decision establishes that causal explanations in high-stakes organizations reflect the conditions of their production [8]. Klein's account of recognition-primed decision making shows that experts under time pressure do not compare alternatives but recognize and act on the first viable option [4] — a cognitive strategy that, in regulated investigation contexts, produces premature convergence precisely when accountability stakes are highest. Star and Strauss show that coordination work performed in the gaps of formal systems becomes invisible [6]: in regulated investigation workflows, the comparative work of holding multiple causal accounts in parallel is unrewarded by closure expectations and absent from final investigation records. Suchman establishes that action is shaped by the immediate conditions of its execution [7], implying that investigation quality is shaped not only by investigator expertise but by the visibility conditions under which reasoning unfolds. Prior CSCW research has examined invisible work and awareness in collaborative settings [2], but has not addressed how comparison collapse in accountability-critical workflows can be made structurally required.

Amershi et al. formalize guidelines for human-AI interaction emphasizing that AI systems should support human control and avoid pre-empting judgment [1]. In regulated environments, this principle takes on particular force: quality decisions must remain with qualified human reviewers. This paper reframes the role of AI from suggesting explanations to enforcing the sustained comparability of alternatives under closure pressure.

3 Method

Both studies were conducted in a pharmaceutical manufacturing quality system operating under GMP regulations, with $N = 20$ participants recruited from a single pharmaceutical quality team with direct experience in deviation investigation, root cause analysis, and CAPA documentation. Participants represented varied levels of experience and roles within the investigative workflow. This sample size is consistent with established think-aloud methodology [5] and reflects the structural constraints of conducting research within a regulated organizational setting. Both studies used the same standardized deviation scenario — an out-of-specification analytical result with sufficient contextual detail to support multiple plausible causal interpretations, including person-centered, procedural, equipment-related, and organizational accounts.

Study 1: Timed Reflection Protocol. Participants produced causal explanations suitable for formal documentation at three timepoints: immediately after reading the scenario (T0), after five minutes of unstructured reflection (T+5), and after ten minutes (T+10). Responses were analyzed for causal breadth, causal type (person-centered, procedural, systemic, or mixed), and evidence of comparison across alternatives.

Study 2: Comparison-Preserving Scaffold. Participants completed the same scenario using an AI-supported reasoning scaffold implemented as a Streamlit web application. The scaffold organized investigation into five stages — case intake, hypothesis expansion, PAC detection, structured comparison, and pre-closure verification — enforcing explicit comparison before closure. The AI component functioned exclusively as a comparison enforcer: detecting convergence signals and triggering reopening prompts, without suggesting causes or generating conclusions. Post-task measures included causal breadth, evidence of explicit comparison, and interaction traces reflecting when and how

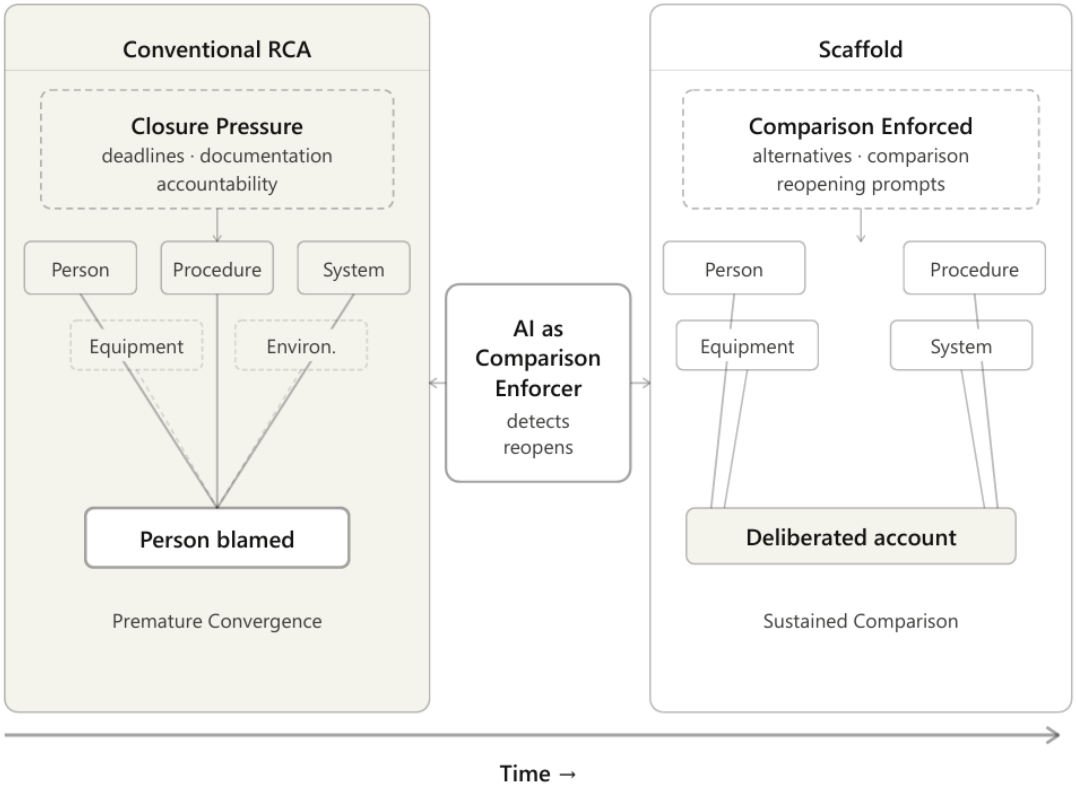

Fig. 1. Premature Accountability Convergence (left) versus a comparison-preserving scaffold (right). Under closure pressure, conventional RCA converges early, collapsing the comparison space before alternatives can be meaningfully evaluated. The scaffold preserves this space by sustaining comparison across multiple alternatives prior to closure.

participants revisited or expanded initial explanations. Post-task discussion probed organizational barriers, particularly tensions between reasoning transparency and existing accountability practices.

4 Findings

4.1 Study 1: Convergence is Structural and Time-Sensitive

At T0, all participants produced single-cause explanations, most person-centered. With additional reflection time, explanations expanded to include procedural, equipment-related, and organizational contributors not present in initial accounts. By T+10, participants either broadened their explanation or became explicitly uncertain about the initial cause — acknowledging contributors they had not yet examined. This pattern establishes the core finding: early convergence reflects constrained investigative conditions, not fixed reasoning capacity. When the workflow defines closure before this broadening can occur, the reasoning state documented is the one produced under maximum constraint.

Post-task discussion revealed a finding more consequential than the timepoint data alone: participants described selecting explanations oriented toward *documentation acceptability* rather than investigative completeness — choosing

contained, person-centered accounts that would satisfy audit review without opening systemic lines of inquiry. Premature convergence in regulated investigations is therefore not only a cognitive phenomenon produced by time pressure; it is a rational strategy under institutional pressure.

4.2 Study 2: Effects of the Comparison-Preserving Scaffold

Under the scaffold, participants produced broader causal accounts and engaged in explicit comparison between alternatives before closure. 18 of 20 participants expanded beyond their initial single-cause explanation, and 15 of 20 reconsidered the case more broadly after scaffolded comparison. Person-centered explanations were less dominant, and final accounts more frequently incorporated procedural and systemic contributors alongside individual factors. In several cases, the scaffold produced a qualitative shift: participants moved from accounts attributing the deviation to an individual actor to accounts identifying the procedural or organizational conditions that made that action possible. Interface usability ratings were consistently positive; the primary adoption barrier was conceptual rather than interactional — the interface was clear, but the idea that investigation reasoning deserves structured support was unfamiliar. Participants questioned whether regulated organizations would deploy a system that makes investigation reasoning more transparent, given that transparency surfaces alternatives that current practice leaves invisible.

5 Discussion

The findings establish that premature convergence is a property of investigative systems rather than individual reasoning — and that it is time-sensitive, and therefore designable. The scaffold demonstrates that structurally enforcing comparison can expand causal breadth and delay premature attribution without automating conclusions or displacing human judgment.

The *comparison enforcer* reframes the role of AI in accountability-critical environments: not to accelerate conclusions but to sustain comparability before attribution. This inverts the dominant model for AI in RCA support, in which suggestion engines and automated classifiers provide anchoring explanations that narrow the comparison space before investigators have explored alternatives independently.

The organizational adoption barrier identified here reveals a deeper tension: making reasoning visible may conflict with existing accountability practices that favor stable, defensible conclusions. A regulated organization that deploys a comparison-preserving scaffold must also accept that its investigation records will document which alternatives were available and not chosen — surfacing conditions that current practice leaves invisible. This suggests that improving investigative reasoning may require redesigning not only tools, but the accountability structures that govern how reasoning is evaluated.

6 Conclusion

Premature convergence in regulated investigation is not a reasoning failure but a property of investigative systems under closure pressure. Comparison collapse is time-sensitive — and therefore designable. An AI-supported scaffold that enforces explicit comparison before closure can expand causal breadth and delay premature attribution. The challenge is not to improve AI accuracy, but to preserve the conditions under which human judgment remains comparative. This requires engaging with the organizational structures that currently make premature convergence rational — a design challenge as much organizational as it is technical.

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